

- Admin *

1) AdminDashboard.jsx :-

```
import Layout from "../components/Layout";

import { Typography } from "@mui/material";

export default function AdminDashboard() {

  return (

    <Layout title="Admin Dashboard">

      <Typography variant="h5">

        System Overview & Controls

      </Typography>

      <ul>

        <li>Manage HR accounts</li>

        <li>View all jobs</li>

        <li>System analytics</li>

      </ul>

    </Layout>

  );

}
```

Api

1) application.api.js :- no code

2) auth.api.js :- no code

3) axios.js :-

```
import axios from "axios";

const api = axios.create({

  baseURL: "http://localhost:5000/api",

});

api.interceptors.request.use((config) => {

  const token = localStorage.getItem("token");

  if (token) {

    config.headers.Authorization = `Bearer ${token}`;

  }

  return config;

});
```

```

});

/* 🔥 AUTO LOGOUT ON 401 */
api.interceptors.response.use(
  (res) => res,
  (err) => {
    if (err.response?.status === 401) {
      localStorage.clear();
      window.location.href = "/login";
    }
    return Promise.reject(err);
  }
);

```

export default api;

4) hr.api.js :- no code

5) job.api.js :- no code

6) offer.api.js :- no code

- Auth

1) Login.jsx :-

```

import { useState, useContext } from "react";
import { useNavigate } from "react-router-dom";
import api from "../api/axios";
import { AuthContext } from "../context/AuthContext";
export default function Login() {
  const [email, setEmail] = useState("");
  const [password, setPassword] = useState("");
  const { login } = useContext(AuthContext);
  const navigate = useNavigate();
  const handleLogin = async () => {
    try {
      const res = await api.post("/auth/login", { email, password });
    }
  }
}

```

```

// save auth
login(res.data);

// role-based redirect
if (res.data.user.role === "HR") {
  navigate("/hr");
} else {
  navigate("/candidate");
}
} catch (err) {
  alert("Invalid credentials");
}
};

return (
  <div>
    <h2>Login</h2>
    <input
      placeholder="Email"
      value={email}
      onChange={e => setEmail(e.target.value)}
    />
    <input
      placeholder="Password"
      type="password"
      value={password}
      onChange={e => setPassword(e.target.value)}
    />
    <button onClick={handleLogin}>Login</button>
  </div>
);
}

```

2) Register.jsx :-

```
import { useState } from "react";
import api from "../api/axios";
export default function Register() {
  const [form, setForm] = useState({
    name: "",
    email: "",
    password: "",
    role: "CANDIDATE"
  });
  const handleChange = (e) => {
    setForm({ ...form, [e.target.name]: e.target.value });
  };
  const handleRegister = async () => {
    await api.post("/auth/register", form);
    alert("Registered successfully. Please login.");
  };
  return (
    <div>
      <h2>Register</h2>
      <input name="name" placeholder="Name" onChange={handleChange} />
      <input name="email" placeholder="Email" onChange={handleChange} />
      <input
        name="password"
        type="password"
        placeholder="Password"
        onChange={handleChange}
      />
      <select name="role" onChange={handleChange}>
        <option value="CANDIDATE">Candidate</option>
        <option value="HR">HR</option>
      </select>
      <button onClick={handleRegister}>Register</button>
    </div>
  );
}
```

```
</div>
```

```
);
```

```
}
```

- Candidate :-

1) ApplyJob.jsx :-

```
import api from "../api/axios";
```

```
export default function ApplyJob({ jobId }) {
```

```
  const apply = async () => {
```

```
    await api.post("/applications/apply", { job_id: jobId });
```

```
    alert("Applied successfully");
```

```
  };
```

```
  return <button onClick={apply}>Apply</button>;
```

```
}
```

2) CandidateDashboard.jsx :-

```
import { useEffect, useState } from "react";
```

```
import api from "../api/axios";
```

```
import Layout from "../components/Layout";
```

```
import {
```

```
  Card, CardContent, Typography,
```

```
  Grid, Button
```

```
} from "@mui/material";
```

```
export default function CandidateDashboard() {
```

```
  const [jobs, setJobs] = useState([]);
```

```
  useEffect(() => {
```

```
    api.get("/jobs").then(res => setJobs(res.data));
```

```
  }, []);
```

```
  const apply = async (job_id) => {
```

```
    await api.post("/applications/apply", { job_id });
```

```
    alert("Applied successfully");
```

```
  };
```

```
  return (
```

```

<Layout title="Candidate Dashboard">
  <Grid container spacing={3}>
    {jobs.map(job => (
      <Grid item xs={12} md={4} key={job.id}>
        <Card>
          <CardContent>
            <Typography variant="h6">{job.title}</Typography>
            <Typography>{job.department}</Typography>
            <Typography>
              ₹{job.salary_min} - ₹{job.salary_max}
            </Typography>
            <Button
              variant="contained"
              sx={{ mt: 2 }}
              onClick={() => apply(job.id)}
            >
              Apply
            </Button>
          </CardContent>
        </Card>
      </Grid>
    )
  )}
</Grid>
</Layout>
);
}

```

3) CandidateProfile.jsx :-

```

import { useState } from "react";
import api from "../api/axios";
export default function CandidateProfile() {
  const [profile, setProfile] = useState({

```

```

    education: "",
    specialization: "",
    experience_years: 0
  });
const handleChange = (e) =>
  setProfile({ ...profile, [e.target.name]: e.target.value });
const saveProfile = async () => {
  await api.post("/candidate", profile);
  alert("Profile created");
};
return (
  <div>
    <h3>Create Profile</h3>
    <input name="education" placeholder="Education" onChange={handleChange} />
    <input
      name="specialization"
      placeholder="Specialization"
      onChange={handleChange}
    />
    <input
      name="experience_years"
      type="number"
      placeholder="Experience"
      onChange={handleChange}
    />
    <button onClick={saveProfile}>Save</button>
  </div>
);
}

```

4) Offer.jsx :-

```
import api from "../api/axios";
```

```

export default function Offer({ offerId }) {
  const respond = async (decision) => {
    await api.post("/offer/respond", {
      offer_id: offerId,
      decision
    });
    alert("Response sent");
  };

  return (
    <>
      <button onClick={() => respond("ACCEPTED")}>Accept</button>
      <button onClick={() => respond("REJECTED")}>Reject</button>
    </>
  );
}

```

5) UploadResume.jsx :-

```

import api from "../api/axios";

export default function UploadResume() {
  const upload = async (e) => {
    const formData = new FormData();
    formData.append("resume", e.target.files[0]);
    await api.post("/resume/upload", formData);
    alert("Resume uploaded");
  };

  return <input type="file" onChange={upload} />;
}

```

- Components :-

1)Layout.jsx :-

```

import { AppBar, Toolbar, Typography, Button, Box } from "@mui/material";

```



```

import { useContext } from "react";
import { AuthContext } from "../context/AuthContext";
import { useNavigate } from "react-router-dom";
export default function Layout({ title, children }) {
  const { logout, user } = useContext(AuthContext);
  const navigate = useNavigate();
  const handleLogout = () => {
    logout();
    navigate("/login");
  };
  return (
    <>
      <AppBar position="static">
        <Toolbar>
          <Typography variant="h6" sx={{ flexGrow: 1 }}>
            AI Recruitment System
          </Typography>
          {/* ===== HR NAV ===== */}
          {user?.role === "HR" && (
            <>
              <Button color="inherit" onClick={() => navigate("/hr")}>
                Dashboard
              </Button>
              <Button color="inherit" onClick={() => navigate("/hr/applications")}>
                Applications
              </Button>
              <Button color="inherit" onClick={() => navigate("/hr/technical-rounds")}>
                Tech Rounds
              </Button>
            </>
          )}
        </Toolbar>
      </AppBar>
    </>
  );
}

```

```

    {/* ===== ADMIN NAV ===== */}

    {user?.role === "ADMIN" && (
      <>
        <Button color="inherit" onClick={() => navigate("/admin")}>
          Admin Dashboard
        </Button>
      </>
    )}

    {/* ===== CANDIDATE NAV ===== */}

    {user?.role === "CANDIDATE" && (
      <>
        <Button color="inherit" onClick={() => navigate("/candidate")}>
          Dashboard
        </Button>
        <Button color="inherit" onClick={() => navigate("/candidate/jobs")}>
          Jobs
        </Button>
        <Button color="inherit" onClick={() => navigate("/candidate/applications")}>
          Applications
        </Button>
      </>
    )}

    <Button color="inherit" onClick={handleLogout}>
      Logout
    </Button>
  </Toolbar>
</AppBar>
<Box sx={{ p: 3 }}>
  <Typography variant="h5" gutterBottom>
    {title}
  </Typography>

```

```

        {children}
      </Box>
    </>
  );
}

```

2) Loader.jsx :-

```

import { CircularProgress, Box } from "@mui/material";
export default function Loader() {
  return (
    <Box sx={{ display: "flex", justifyContent: "center", mt: 4 }}>
      <CircularProgress />
    </Box>
  );
}

```

3) NotFound.jsx :-

```

export default function NotFound() {
  return (
    <div style={{ padding: 40 }}>
      <h2>404 – Page Not Found</h2>
      <p>The page you're looking for does not exist.</p>
    </div>
  );
}

```

- Context :-

1) AuthContext.jsx :-

```

import { createContext, useState, useEffect } from "react";
import { jwtDecode } from "jwt-decode";
export const AuthContext = createContext();
export function AuthProvider({ children }) {
  const [user, setUser] = useState(null);

```

```

const [token, setToken] = useState(null);
const [loading, setLoading] = useState(true);
useEffect(() => {
  try {
    const storedToken = localStorage.getItem("token");
    if (storedToken) {
      const decoded = jwtDecode(storedToken);
      setUser(decoded);
      setToken(storedToken);
    }
  } catch (err) {
    console.error("Invalid token");
    localStorage.clear();
  } finally {
    setLoading(false);
  }
}, []);
const login = (data) => {
  localStorage.setItem("token", data.token);
  const decoded = jwtDecode(data.token);
  setUser(decoded);
  setToken(data.token);
};
const logout = () => {
  localStorage.clear();
  setUser(null);
  setToken(null);
};
if (loading) return null;
return (
  <AuthContext.Provider value={{ user, token, login, logout }}>

```

```

        {children}
      </AuthContext.Provider>
    );
  }

```

- Hr

1)ApplicationList.jsx :-

```

import { useEffect, useState } from "react";
import api from "../api/axios";
import Layout from "../components/Layout";
import {
  Table, TableBody, TableCell, TableContainer,
  TableHead, TableRow, Paper, Button, Chip
} from "@mui/material";
const statusColor = (status) => {
  switch (status) {
    case "TECHNICAL_ROUND": return "primary";
    case "REJECTED": return "error";
    case "HIRED": return "success";
    default: return "default";
  }
};

export default function ApplicationList() {
  const [apps, setApps] = useState([]);
  useEffect(() => {
    api.get("/hr/applications").then(res => setApps(res.data));
  }, []);

  const decision = async (id, decision) => {
    await api.post("/hr/decision", { application_id: id, decision });
    window.location.reload();
  };

  return (

```

```

<Layout title="Applications">
  <TableContainer component={Paper}>
    <Table>
      <TableHead>
        <TableRow>
          <TableCell>Candidate</TableCell>
          <TableCell>Job</TableCell>
          <TableCell>Status</TableCell>
          <TableCell>Action</TableCell>
        </TableRow>
      </TableHead>
      <TableBody>
        {apps.map(app => (
          <TableRow key={app.id}>
            <TableCell>{app.Candidate.User.name}</TableCell>
            <TableCell>{app.Job.title}</TableCell>
            <TableCell>
              <Chip
                label={app.status}
                color={statusColor(app.status)}
              />
            </TableCell>
            <TableCell>
              <Button
                variant="contained"
                color="success"
                size="small"
                onClick={() => decision(app.id, "SHORTLISTED")}
              >
                Shortlist
              </Button>
            </TableCell>
          </TableRow>
        ))}
      </TableBody>
    </Table>
  </TableContainer>
</Layout>

```

```

        &nbsp;
        <Button
            variant="contained"
            color="error"
            size="small"
            onClick={() => decision(app.id, "REJECTED")}
        >
            Reject
        </Button>
    </TableCell>
</TableRow>
    )}
</TableBody>
</Table>
</TableContainer>
</Layout>
);
}

```

2) HRDashboard.jsx :-

```

import { useEffect, useState } from "react";
import api from "../api/axios";
import Layout from "../components/Layout";
import { Grid, Card, CardContent, Typography } from "@mui/material";
import Loader from "../components/Loader";
export default function HRDashboard() {
    const [stats, setStats] = useState(null);
    const [loading, setLoading] = useState(true);
    useEffect(() => {
        const fetchDashboard = async () => {
            try {
                const res = await api.get("/hr/dashboard");
            }

```

```

        setStats(res.data);
    } catch (err) {
        console.error("Failed to load dashboard", err);
    } finally {
        setLoading(false);
    }
};

fetchDashboard();
}, []);

if (loading) return <Loader />;

const CardItem = ({ title, value }) => (
    <Card>
        <CardContent>
            <Typography variant="h6">{title}</Typography>
            <Typography variant="h4">{value ?? 0}</Typography>
        </CardContent>
    </Card>
);

return (
    <Layout title="HR Dashboard">
        <Grid container spacing={3}>
            <Grid item xs={12} md={3}>
                <CardItem
                    title="Total Applications"
                    value={stats.totalApplications}
                />
            </Grid>
            <Grid item xs={12} md={3}>
                <CardItem
                    title="Shortlisted"
                    value={stats.shortlisted}

```



```

    />
  </Grid>
  <Grid item xs={12} md={3}>
    <CardItem
      title="Rejected"
      value={stats.rejected}
    />
  </Grid>
  <Grid item xs={12} md={3}>
    <CardItem
      title="Completed Rounds"
      value={stats.completedTechRounds}
    />
  </Grid>
</Grid>
</Layout>
);
}

```

3)MalpracticeSummary.jsx :-

```

import { useEffect, useState } from "react";
import api from "../api/axios";
import Layout from "../components/Layout";
import { List, ListItem, Chip } from "@mui/material";
export default function MalpracticeSummary({ applicationId }) {
  const [events, setEvents] = useState([]);
  useEffect(() => {
    api.get(`/malpractice/summary/${applicationId}`)
      .then(res => setEvents(res.events));
  }, []);
  return (
    <Layout title="Malpractice Summary">

```

```

    {events.map(e => (
      <List key={e.id}>
        <ListItem>
          <Chip
            label={e.type}
            color={e.severity >= 3 ? "error" : "warning"}
          />
          &nbsp; Severity: {e.severity}
        </ListItem>
      </List>
    ))}
  </Layout>
);
}

```

- Interview :-

Interview.jsx :-

```

import { useEffect, useRef } from "react";
import Layout from "../components/Layout";

```

```

export default function Interview() {
  const videoRef = useRef();

  useEffect(() => {
    navigator.mediaDevices.getUserMedia({ video: true, audio: true })
      .then(stream => {
        videoRef.current.srcObject = stream;
      });
  }, []);

  return (
    <Layout title="AI Technical Interview">

```

```
<video ref={videoRef} autoPlay width="400" />
```

```
<h3>Question:</h3>
```

```
<p>Explain REST vs GraphQL</p>
```

```
<textarea rows={5} placeholder="Type your answer..." />
```

```
<button>Submit Answer</button>
```

```
</Layout>
```

```
);
```

```
}
```

- Mcq :-

1)MCQTest.jsx :-

```
import { useEffect, useState } from "react";
```

```
import api from "../api/axios";
```

```
import Timer from "../Timer";
```

```
import Layout from "../components/Layout";
```

```
export default function MCQTest({ applicationId }) {
```

```
  const [questions, setQuestions] = useState([]);
```

```
  const [answers, setAnswers] = useState({});
```

```
  const [submitted, setSubmitted] = useState(false);
```

```
  useEffect(() => {
```

```
    api.get(`/mcq/test/${applicationId}`)
```

```
      .then(res => setQuestions(res.data.questions));
```

```
  }, []);
```

```
/* 🚧 TAB SWITCH DETECTION */
```

```
useEffect(() => {
```

```
  const handleBlur = () => {
```

```
    api.post("/malpractice/log", {
```

```
      application_id: applicationId,
```

```
      type: "TAB_SWITCH",
```

```

    severity: 3
  });
};
window.addEventListener("blur", handleBlur);
return () => window.removeEventListener("blur", handleBlur);
}, []);
const submitTest = async () => {
  if (submitted) return;
  setSubmitted(true);
  await api.post("/mcq/submit", {
    application_id: applicationId,
    answers
  });
  alert("Test submitted");
};
return (
  <Layout title="MCQ Test">
    <Timer minutes={30} onTimeUp={submitTest} />
    {questions.map((q, index) => (
      <div key={q.id}>
        <h4>{index + 1}. {q.question}</h4>
        {[ "A", "B", "C", "D" ].map(opt => (
          <label key={opt}>
            <input
              type="radio"
              name={q.id}
              onChange={() =>
                setAnswers({ ...answers, [q.id]: opt })
              }
            />
            {q[`option_${opt.toLowerCase()}`]}

```

```

        </label>

    )))
</div>

)))
<button onClick={submitTest}>Submit</button>
</Layout>

);
}

```

2) Timer.jsx :-

```

import { useEffect, useState } from "react";
export default function Timer({ minutes, onTimeUp }) {
  const [time, setTime] = useState(minutes * 60);
  useEffect(() => {
    if (time <= 0) {
      onTimeUp();
      return;
    }
    const interval = setInterval(() => {
      setTime((t) => t - 1);
    }, 1000);
    return () => clearInterval(interval);
  }, [time]);
  return (
    <h3>
      Time Left: {Math.floor(time / 60)}:{time % 60}
    </h3>
  );
}

```

- Routes

1)ProtectedRoutes.jsx:-

```

import { Navigate } from "react-router-dom";

```

```

import { useContext } from "react";
import { AuthContext } from "../context/AuthContext";
export default function ProtectedRoute({ children, role }) {
  const { user } = useContext(AuthContext);
  if (!user) {
    return <Navigate to="/login" replace />;
  }
  if (role && user.role !== role) {
    return <Navigate to="/" replace />;
  }
  return children;
}

```

- App.jsx :-

```

import { Routes, Route, Navigate } from "react-router-dom";
import Login from "../auth/Login";
import ProtectedRoute from "../routes/ProtectedRoute";

import HRDashboard from "../hr/HRDashboard";
import CandidateDashboard from "../candidate/CandidateDashboard";
import AdminDashboard from "../admin/AdminDashboard";

export default function App() {
  return (
    <Routes>
      <Route path="/login" element={<Login />} />

      {/* HR */}
      <Route
        path="/hr"
        element={
          <ProtectedRoute role="HR">
            <HRDashboard />
          </ProtectedRoute>
        }
      />

      {/* Candidate */}
      <Route
        path="/candidate"

```

```

    element={
      <ProtectedRoute role="CANDIDATE">
        <CandidateDashboard />
      </ProtectedRoute>
    }
  />

  {/* Admin */}
  <Route
    path="/admin"
    element={
      <ProtectedRoute role="ADMIN">
        <AdminDashboard />
      </ProtectedRoute>
    }
  />

  {/* Default */}
  <Route path="*" element={<Navigate to="/login" />} />
</Routes>
);
}

```

- Main.jsx :

```

• import React from "react";
• import ReactDOM from "react-dom/client";
• import App from "./App";
• import { BrowserRouter } from "react-router-dom";
• import { AuthProvider } from "../context/AuthContext";
• import { ThemeProvider } from "@mui/material/styles";
• import CssBaseline from "@mui/material/CssBaseline";
• import theme from "../theme";
•
• ReactDOM.createRoot(document.getElementById("root")).render(
•   <BrowserRouter>
•     <ThemeProvider theme={theme}>
•       <CssBaseline />
•       <AuthProvider>
•         <App />
•       </AuthProvider>
•     </ThemeProvider>
•   </BrowserRouter>
• );
•

```

- Theme.jsx :-

```

• import { createTheme } from "@mui/material/styles";

```

```

•
•   const theme = createTheme({
•     palette: {
•       primary: {
•         main: "#1976d2", // corporate blue
•       },
•       secondary: {
•         main: "#2e7d32", // green
•       },
•     },
•     typography: {
•       fontFamily: "Roboto, sans-serif",
•     },
•   });
•
•   export default theme;
•

```

index.html :-

```

<!-- frontend/index.html -->

<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <title>Mask Polymers</title>

  </head>

  <body>

    <div id="root"></div>

    <script type="module" src="/src/main.jsx"></script>

  </body>

</html>

```

Package.json :-

```

{
  "name": "mask-polymers-frontend",

```



```
"version": "1.0.0",
"private": true,
"type": "module",
"scripts": {
  "dev": "vite",
  "build": "vite build",
  "preview": "vite preview"
},
"dependencies": {
  "axios": "^1.6.7",
  "jwt-decode": "^4.0.0",
  "react": "^18.2.0",
  "react-dom": "^18.2.0",
  "react-router-dom": "^6.21.0"
},
"devDependencies": {
  "@vitejs/plugin-react": "^4.2.0",
  "vite": "^5.0.0"
}
}
```

Vite.config.js :-

```
import { defineConfig } from "vite";
import react from "@vitejs/plugin-react";

export default defineConfig({
  plugins: [react()],
});
```

